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Blood vessels provided the template for vertebral and costal pneumatization in sauropod dinosaurs

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Pneumatization of the postcranial skeleton, especially the vertebral column, was a key innovation in the evolution of gigantism in sauropod dinosaurs. Despite a century and a half of study, the morphogenetic rules that govern skeletal pneumatization in sauropods — and indeed in extant birds — remain obscure. It has been suggested that pneumatic diverticula of the respiratory system follow pre-existing arteries and veins in pneumatizing the skeleton. This could explain why skeletal pneumaticity is so variable, both in birds and in their extinct outgroups, but published evidence for the “blood vessels first” hypothesis is scant. In extant amniotes, arteries and veins enter and leave the vertebrae and ribs in seven locations: (1) outer surfaces of the neural arch, (2) roof and walls of the neural canal, (3) floor of the neural canal, (4) ventral surfaces of the transverse processes and inner surfaces of the ribs, (5) lateral surfaces of the centrum, (6) ventral surface of the centrum, and (7) medial surfaces of the haemal arch. Vascular foramina are present in all of these locations in apneumatic vertebrae of sauropods, suggesting a highly conserved pattern of vertebral vascularization. Pneumatic fossae and foramina are also present in all of these locations in at least some vertebrae and ribs of sauropods, though not all locations may be pneumatized in a single individual, or even across a clade. The striking similarity of vertebral vascularization and pneumatization in sauropods is consistent with the hypothesis that pneumatic diverticula followed blood vessels, and is difficult to explain any other way. The same relationship of pneumatic features to blood vessels probably existed in non-avian theropods and pterosaurs, but has not yet been documented. The relationships of vascular and pneumatic features in sauropods and other extinct archosaurs could be elucidated by systematic surveys of existing museum collections.

Blood vessels provided the template for vertebral and costal pneumatization in sauropod dinosaurs

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Abstract

Pneumatization of the postcranial skeleton, especially the vertebral column, was a key innovation in the evolution of gigantism in sauropod dinosaurs. Despite a century and a half of study, the morphogenetic rules that govern skeletal pneumatization in sauropods — and indeed in extant birds — remain obscure. It has been suggested that pneumatic diverticula of the respiratory system follow pre-existing arteries and veins in pneumatizing the skeleton. This could explain why skeletal pneumaticity is so variable, both in birds and in their extinct outgroups, but published evidence for the "blood vessels first" hypothesis is scant. In extant amniotes, arteries and veins enter and leave the vertebrae and ribs in seven locations: (1) outer surfaces of the neural arch, (2) roof and walls of the neural canal, (3) floor of the neural canal, (4) ventral surfaces of the transverse processes and inner surfaces of the ribs, (5) lateral surfaces of the centrum, (6) ventral surface of the centrum, and (7) medial surfaces of the haemal arch. Vascular foramina are present in all of these locations in apneumatic vertebrae of sauropods, suggesting a highly conserved pattern of vertebral vascularization. Pneumatic fossae and foramina are also present in all of these locations in at least some vertebrae and ribs of sauropods, though not all locations may be pneumatized in a single individual, or even across a clade. The striking similarity of vertebral vascularization and pneumatization in sauropods is consistent with the hypothesis that pneumatic diverticula followed blood vessels, and is difficult to explain any other way. The same relationship of pneumatic features to blood vessels probably existed in non-avian theropods and pterosaurs, but has not yet been documented. The relationships of vascular and pneumatic features in sauropods and other extinct archosaurs could be elucidated by systematic surveys of existing museum collections.

Introduction

Pneumatization of the vertebral column was a key innovation in the evolution of gigantism in sauropod dinosaurs (Sander et al. 2010, Sander 2013). Pneumatic features in sauropods vary significantly among taxa (Wedel 2003), among individuals (Wedel and Taylor 2013), within the vertebral column of an individual (Wilson 2012), and even between the left and right sides of a single individual (Wilson 1999, Taylor and Naish 2007, Zurriaguz and Alvarez 2014, Tschopp and Upchurch 2019, Moore et al. 2020, Zurriaguz 2025), and this variation is consistent with that observed in pneumaticity in extant birds (O'Connor 2006). In extant birds, pneumatic diverticula have been reported to follow pre-existing blood vessels in pneumatizing the bones of the postcranial skeleton (Bremer 1940:200). Blood vessels are highly variable among and within individuals, and to the extent that pneumatic diverticula follow blood vessels, pneumatization will likewise be variable (Taylor and Wedel 2021).

In the present paper we explore the locations where vascular and pneumatic features appear on vertebrae and the reasons why they appear in these particular locations.

Anatomical nomenclature

The term “transverse process” properly refers to processes projecting laterally (or dorsolaterally, posterolaterally, etc.) from the neural arch. The transverse process bears the diapophysis with which the tuberculum of the rib articulates. The term is sometimes also used for the processes that extend laterally from caudal vertebrae. In most caudal vertebrae these processes are actually coalesced costal elements (caudal ribs; Fig. 1). Gallina and Otero (2009) argued that the dorsoventrally tall, plate-like lateral processes of proximal caudal vertebrae in some sauropods consist of a combination of true transverse processes (parts of the neural arches) and fused caudal ribs. Caudal vertebrae otherwise do not have true transverse processes, which are found only in dorsal vertebrae (and in cervicals, although since they are fused to the cervical ribs they are rarely referred to as separate parts of cervical vertebrae). For clarity, we use the term “transverse process” only for the diapophysis-bearing portion of the neural arch, and not for costal elements.

Foramina in the walls of bones that transmit blood vessels have historically been referred to as nutrient foramina, vascular foramina, or neurovascular foramina. Blood vessels are always innervated by microscopic nerve fibers that provide sensory feedback and regulate vasodilation and vasoconstriction (nervi vasorum, see Standring 2008:138), and the living tissue of bones is innervated by additional nerve fibers that accompany blood vessels as they pass inside the bones (Hurrell 1937, Steverink et al. 2021), so technically all vascular foramina are in fact neurovascular. Similarly, all macroscopic nerves are accompanied by microscopic (and sometimes macroscopic) blood vessels (vasa nervorum, see Adams 1942, El-Barrany et al. 1999, and Standring 2008:53), so technically all nervous foramina are also neurovascular. Any given foramen that is said to transmit an artery, vein, or nerve, actually transmits all three, at least at the microscopic level. Gailloud (2021) found that in humans the basivertebral foramina – paired foramina in the floor of the neural canal that admit the basivertebral veins into the dorsal surface of the centrum – also admit small arteries into the centrum along with the veins. Still, it is often the case that vessels that pass into bones are accompanied only by microscopic nerves, or vice versa, so we use the term “vascular foramina” for those whose volume is mostly occupied by blood vessels.

Similarly, O'Connor (2006) found in dissecting birds that had been treated with vascular injections that the pneumatic foramina transmitted blood vessels along with pneumatic diverticula. This probably accounts for the scarcity of purely neurovascular foramina at the macroscopic level in bones that have been completely pneumatized, although neurovascular foramina are sometimes found next to and even inside pneumatic fossae in partially-pneumatized elements (e.g., Taylor and Wedel 2021: figure 6). So in the same sense that vascular and nervous foramina are all technically neurovascular, pneumatic foramina are often and possibly always neurovascular as well. For simplicity and to maintain consistency with existing literature, we refer to foramina that transmit pneumatic diverticula as "pneumatic foramina", but the fact that such foramina also transmit blood vessels will be important in the discussion below.

Museum Abbreviations

- AMNH — American Museum of Natural History, New York, New York, USA.
- BIBE — Big Bend National Park, held at Perot Museum of Nature and Science, Dallas, Texas, USA.
- BYU — Brigham Young University Museum of Paleontology, Provo, Utah, USA.
- CM — Carnegie Museum of Natural History, Pittsburgh, Pennsylvania, USA.
- FHPR — Utah Field House of Natural History State Park Museum, Vernal, Utah, USA.
- FMNH — Field Museum of Natural History, Chicago, Illinois, USA.
- IVPP — Institute of Vertebrate Paleontology and Paleoanthropology, Beijing, China.
- LACM — Natural History Museum of Los Angeles County, Los Angeles, USA.
- MACN — Museo Argentino de Ciencias Naturales "Bernardino Rivadavia", Buenos Aires, Argentina.
- MB — Humboldt Museum für Naturkunde Berlin, Berlin, Germany.
- NSMT — National Science Museum, Tokyo, Japan.
- RRBP — Rukwa Rift Basin Project, Tanzanian Antiquities Unit, Dar es Salaam, Tanzania.
- YPM — Yale Peabody Museum, New Haven, Connecticut, USA.

Description

Vertebral articulation

Virtually all tetrapod vertebrae are composed of the same elements: the centrum, neural arch, haemal arch (haemal arch), and paired costal elements (Fig. 1). Not all elements are present in all vertebrae, and the costal elements may be fused (as is generally the case in the neck, sacrum and tail) or mobile (as is generally the case in the trunk), but this is the general plan.

[Figure 1 goes here]

Vertebral vascularization

The vertebral elements are served by a system of arteries and veins. Specific branching patterns are variable among lineages and individuals, but the basic plan is highly conserved. In amniotes, early development of the spine and spinal cord is supported by bilaterally paired intersegmental arteries that branch from the dorsal aorta and pass between adjacent somites (Moran et al. 2012, Moore et al. 2015). After resegmentation, each pair of arteries lies alongside a vertebral centrum, and they become the paired segmental arteries that serve each vertebral segment and its associated muscles and neural tissues (e.g., intercostal arteries and lumbar arteries). The branching of paired segmental arteries from the dorsal aorta produces a ladder-like appearance; this is beautifully illustrated in contrast-enhanced radiographs of newly-hatched chickens (Levinsohn et al. 1984).

To the best of our knowledge, there is no broad, phylogenetically-grounded synthesis of blood supply to the vertebrae in amniotes. Vascular supply to the vertebrae is best understood in mammals (Amato et al. 1959, Crock 1960, Crock et al. 1973, Smuts 1975, Crock and Yoshizawa 1976, Ratcliffe 1982, Konerding and Blank 1987, Gailloud 2019, 2021, and Travan et al. 2015), but aspects of the general pattern described below have been documented in most other amniotes. Where some clades diverge from this pattern, it is generally in the loss of vessels in one or more of these areas. For example, Amato et al. (1959) reported that in rabbits no large vessels entered the external surfaces of the vertebral centra, which were supplied entirely by blood vessels within the neural canal. In contrast, external vascular foramina are common in the centra of many lizards (Camp 1923, Mookerjee and Das 1933). In geckos, vascular foramina are prominent on the ventral centrum but not the lateral surfaces of the centrum, and foramina are occasionally present on the neural arches as well (Holder 1960). In rare cases, vascular traces on vertebrae allow specific vessels to be reconstructed even in extinct taxa, like the intercostal vessels identified in Mesozoic crocodilians by Gunther (1924).

In birds, vascular foramina are often present on the lateral surfaces of the centra, at least where they have not been overprinted by pneumatization (O'Connor 2006, Taylor and Wedel 2021). The free (unfused) caudal vertebrae and pygostyle bear ventral grooves for the caudal aorta and caudal veins (Rashid et al. 2018, Imai et al. 2019). Blood supply to the spinal cord in birds is broadly similar to that of mammals and other animals, by segmental spinal arteries that branch from the segmental arteries (Roncali and Bosco 1980, Hasouna et al. 1995a, b). In contrast to mammals and most other amniotes, birds lack basivertebral veins, and instead have an expansive internal venous plexus within the neural canal (Burger and Estavillo 1977, O'Connor 2006). Even pneumatic vertebrae of birds may have small vascular foramina within the neural canal (Atterholt et al. 2025).

[Figure 2 goes here]

Figure 2 shows a generalized schema of vertebral vascularization in amniotes, based on the literature cited above. Seven distinct groups of arteries and veins enter or exit the vertebrae at distinctive locations (Table 1): (1) the neural arch; (2) dorsal roof or lateral walls of the neural canal (neural arch elements); (3) ventral floor of the neural canal (centrum); (4) ventral surfaces of the transverse processes, or inner surfaces of the costal elements; (5) lateral surfaces of the centrum; (6) ventral surface of the centrum; (7) medial surface of the haemal arch (Fig. 3).

[Figure 3 goes here]

We assembled this seven-site model of vertebral vascularization based on published accounts, and tested it by documenting vascular foramina in the vertebrae of extant taxa that phylogenetically bracket sauropod dinosaurs: the American alligator, *Alligator mississippiensis*, a pseudosuchian archosaur in which postcranial pneumaticity is primitively absent, and the Emperor penguin, *Aptenodytes forsteri*, a flightless diving bird in which postcranial pneumaticity is secondarily lost. We chose *Alligator* for easy availability of skeletal material, and *Aptenodytes* because of its size (meaning that vascular foramina would be larger and easier to see and photograph) and lack of pneumaticity (meaning that vascular foramina would be identifiable as such, because they had not been overprinted by pneumatization).

[Figure 4 goes here]

We found vascular foramina in all seven of the predicted sites in *Alligator* (Fig. 4), and did not observe prominent vascular foramina in any additional sites. In *Aptenodytes*, we found vascular foramina in the six vertebral sites (Fig. 5). Vascularization of the haemal arches cannot be assessed in birds, because haemal arches were lost in the evolution of birds from other dinosaurs (Rashid et al. 2014). The caudal artery (= caudal aorta) is still present in extinct birds, and it terminates distally by entering the pygostyle through a foramen on the ventral surface (Imai et al. 2019).

[Figure 5 goes here]

We also found vascular foramina at all seven of the predicted sites in the vertebrae of sauropod dinosaurs (Fig. 6). The precaudal vertebrae of most neosauropods are so extensively pneumatized that finding vascular foramina that can be clearly distinguished from pneumatic foramina is challenging. Proximal caudal vertebrae are also pneumatized in many sauropod lineages (Wedel and Taylor 2013, Beeston et al. 2026), so we focused on taxa in which caudal pneumaticity is limited or nonexistent. FHPR 13443, a caudal vertebral centrum from the Upper Jurassic Morrison Formation of Utah that is probably referrable to the basal macronarian *Camarasaurus*, bears vascular foramina on the lateral and ventral surfaces of the centrum, and in the floor of the neural canal (dorsal surface of the centrum). The caudal vertebrae of USNM 1556, a partial skeleton of the titanosaur *Alamosaurus* from the Upper Cretaceous North Horn Formation of Utah, bear vascular foramina on the lateral and ventral surfaces of the centra, the lateral aspects of the neural spines, the ventral surfaces of the costal processes, and the medial surfaces of the haemal arches (the neural canals have not been prepped in these vertebrae). Small vascular foramina are visible in the roof of the neural canal in the unfused neural arches of CM 555, a partial skeleton of *Brontosaurus* from the Upper Jurassic Morrison Formation of Wyoming.

[Figure 6 goes here]

Pneumatic sites in sauropod vertebrae

The seven sites where arteries and veins enter and exit the bones are also the sites where the vertebral elements may be pneumatized by diverticula of the respiratory system. In the skeletons of sauropods, pneumatic fossae and foramina have been documented at all seven of these locations (Fig. 7, Table 1), although not all locations may be pneumatized in a single individual, or even across a clade.

[Figure 7 goes here]

The most obvious and best-documented pneumatic traces in sauropod vertebrae are the fossae and foramina on the lateral aspects of the neural arch and centrum (e.g., Wedel 2005:figure 7.2). Also widespread are peduncular fossae: the paired pneumatic fossae or foramina on the anterior or posterior aspect of the neural arch, lateral to the neural canal. These are found in pterosaurs, sauropods, and non-avian theropods, as illustrated and discussed by Taylor and Wedel (2021:figure. 4). These probably also represent pneumatization of the neural arch by diverticula following the segmental spinal arteries that serve the spinal cord.

Pneumatic openings from the neural canal into the neural arch and centrum are less well-documented, in part because the inner surface of the neural canal is difficult to prepare or observe, especially in proportionally long vertebrae. Nevertheless, a few examples of pneumatic fossae and foramina inside the neural canal have been described, both from CT scans (Schwarz and Fritsch 2006:figure 4H–J; Aureliano et al. 2021:figure 2.3) and from direct observation in broken or unfused specimens (Atterholt and Wedel 2018, Gorscak and O'Connor 2019). The centrodiapophyseal fossae of sauropod presacral vertebrae (Wilson et al. 2011) represent pneumatization of the ventral surfaces of the transverse processes and adjacent regions of the neural arches, presumably by diverticula that followed segmental arteries (= intercostal arteries in the dorsal series). The proximal portions of the dorsal ribs are pneumatized in the mamenchisaurid *Xinjiangtitan* (Zhang et al. 2022), a few diplodocids (Lovelace et al. 2008:figure 9, King et al. 2024) and in most titanosauriforms (e.g., Janensch 1950:figure 107, Gomani 2005:figure 12, Curry Rogers 2009:figure 30, Taylor and Wedel 2023).

Pneumatization of the ventral surface of the centrum has been documented in multiple regions of the vertebral column. In cervical vertebrae, they are found in the holotype of *Mamenchisaurus sinocanadorum* IVPP V10603 (Moore et al. 2023), the referred *Apatosaurus ajax* specimen NSMT-PV 20375 (Upchurch et al. 2005:plate 1K), the Wyoming *Supersaurus* (Lovelace et al. 2008:figure 4A) and the *Giraffatitan brancai* lectotype MB.R.2180 (personal observation) and paralectotype MB.R.2181 (Janensch 1950::figures 40, 42) among others. We have not observed pneumatic features on ventral surfaces of dorsal or sacral vertebrae, despite having looked for them. In caudal vertebrae, ventral pneumaticity is seen in, for example, *Rocasaurus muniozi* MPCA- Pv 54 (Cerda et al. 2012:figure 1G), an indeterminate saltasaurine titanosaur (Zurriaguz et al. 2017:figure 5), and the diplodocoid BYU 11505 (pers. obs.).

Ventral pneumatization may be more common than currently appreciated due to the difficulty in studying the ventral aspects of large, fragile vertebrae that are often stored upright on bases (see, e.g., Taylor 2022: figures 5 and 11). This is a rare situation in which mounted sauropod skeletons can be more rather than less scientifically useful than individual elements in collections: they provide an opportunity to view the ventral aspects of large vertebrae, at least in those cases where the ventral aspects are not blocked by supporting armatures.

Finally, pneumatization of the haemal arches has been documented in the indeterminate saltasaurine titanosaur MACN-Pv RN 233-4 (Zurriaguz et al. 2017:figure 9). As is often the case with pneumatic features (Wedel and Taylor 2013) it seems possible that this feature also exists in other known elements, but has not been identified because it was not expected. More careful inspection of other sauropod haemal arches may present a fruitful opportunity to extend our knowledge in this area.

Discussion

Do pneumatic diverticula follow blood vessels?

In describing the process of pneumatization of the humerus in the chicken, *Gallus domesticus*, Bremer (1940: 199) argued that having a pneumatic diverticulum ('air sac' of his usage) in contact with a bone was necessary but not sufficient to lead to internal pneumatization of the bone:

> The lower bay of the air sac meets the periosteum, the connective tissue of the sac fuses with it, and the sac flattens against the bone, but does not enter it. Instead, the sac wall becomes wrinkled, the epithelium thickens, and new side pockets burrow between the muscles at some distance from the bone, seeking new regions in which the sac may expand. The apposition of the air sac to the bone cortex is not in itself sufficient to ensure entrance into the bone.

Instead of burrowing directly through the cortical bone, the diverticulum induces bone resorption (through a process of cell signalling that is still not well understood) only in the vicinity of pre-existing blood vessels. As described by Bremer (1940):

> The air sac, above the teres major [muscle], meets the bone at the level of the exit of one of the veins, in the region of the newly formed cortex. [...] The inner layer of the periosteum is replaced by a delicate tissue closely resembling mesenchyma. These changes are at first limited to the area immediately surrounding the vein exit (fig. l), but they extend inward chiefly along the vein, with the result that the latter soon lies in a constantly enlarging tubular compartment,

> surrounded by bone trabeculae, filled by mesenchyma… Into this loose tissue, along the vein, the air sac finally grows in the form of a long tube. [...] The actual entrance of the air sac into the main marrow cavity is effected at first at the internal opening of the vein.

To our knowledge, this vivid description of the process of pneumatization is still the most detailed available for any postcranial element. (Cranial pneumatization is better studied but not really comparable, since the paranasal and paratympanic diverticula [sensu Witmer 1997] are in intimate contact with the bones of the skull from their earliest stages of development.)

As detailed as Bremer's account is, it is still just one study on the pneumatization of one element — the humerus — in a single species of bird. But other lines of evidence corroborate the hypothesis that pneumatic diverticula follow blood vessels during the process of pneumatizing the postcranial skeleton:

1. Injection studies show that 'pneumatic' foramina are often co-occupied by both pneumatic diverticula and blood vessels (O'Connor 2006:1208-1209).
2. In ducks, *Anas platyrhynchos*, not only do pneumatic fossae on the lateral aspects of the dorsal vertebrae form in places where vascular foramina are already present, but vascular foramina remain visible within the pneumatic fossae (Taylor and Wedel 2021:figure 6).
3. At the macroscopic level, purely vascular foramina are few to nonexistent in well-pneumatized elements. This is certainly true in birds. The cortical bone of sauropod vertebrae is so thick that tiny vascular foramina are often visible on the outer surfaces of pneumatic vertebral centra, especially ventrally and adjacent to the articular surfaces, but obvious vascular foramina are absent on the portions of the centra and especially the neural arches that have been remodeled by pneumaticity. This is consistent with the hypothesis that vascular foramina tend to become overprinted by pneumatic diverticula.

If blood vessels and pneumatic diverticula co-occupy the same foramina, can we say which anatomical system was present first? The vascular network of birds is substantially complete before hatching, including the segmental arteries that supply the vertebrae (Levinsohn et al. 1984). Indeed, bones cannot ossify or be remodeled without substantial blood supply, and vessels come to occupy neurovascular foramina after bone ossifies around them (Seymour et al. 2012, Hendriks and Rasamany 2020). In contrast, pneumatization of the postcranial skeleton doesn't begin until several weeks into post-hatching ontogeny (Hogg 1984). Even absent Bremer's (1940) description of pneumatic diverticula following blood vessels and exploiting existing neurovascular foramina to gain access to the interiors of postcranial bones, we would predict this based on the facts that (1) blood vessels and pneumatic diverticula co-occupy the same foramina, and (2) blood vessels develop before pneumatic diverticula.

In extant birds, pneumatic foramina can occur on the anterior or posterior surfaces of the proximal femur (Livesey and Zusi 2006). Birds have femoral circumflex arteries that encircle the proximal femur (Baumel 1993), and presumably pneumatic diverticula follow these arteries. It would be interesting to know whether anterior versus posterior pneumatization of the femur correlates with greater blood supply to the femur on its anterior or posterior surfaces; indeed, this would make a near-perfect test of the hypothesis that pneumatic diverticula follow blood vessels.

Although further experimental validation would of course be welcome, the hypothesis that pneumatic diverticula follow blood vessels and that pneumatic foramina are created by the remodeling and expansion of vascular foramina is consistent with all available evidence.

Sites of vascularization are sites of pneumatization

In the present study, we have shown that sauropods manifest pneumatic fossae or foramina in all seven of the sites where blood vessels penetrate vertebrae, corroborating the hypothesis that pneumaticity followed vascularization. As discussed by Taylor and Wedel (2021), this developmental pattern explains why bone — usually the least variable material in the vertebrate body — shows so much variation in its pneumatic morphology: pneumatization follows one of the most developmentally labile systems of the body (see Berger 1956 on the relative variability of bones, blood vessels, and other body systems).

The same relationship of pneumatic features to blood vessels probably existed in other animals with postcranial pneumaticity: non-avian theropods and pterosaurs. The presence of pneumatic features at all seven sites of vascularization has not yet been documented in those clades, but since the known sites of pneumatization are analogous to those documented here for sauropods (e.g. O'Connor 2006, Claessens et al. 2009, Benson et al. 2012), it would not be surprising if pneumatic features were to be found in all sites within both clades.

Pneumatic sites unrelated to major blood vessels

The schema presented here does not explain all known pneumatization in sauropods. For example, according to the seven-sites plan outlined above, pneumatic openings in the ribs should be found only on their posteromedial aspects, corresponding to the location of the intercostal arteries. And indeed in the great majority of documented pneumatic ribs, this is the case. However, there are exceptions. For example, right dorsal rib 2 of the well-preserved *Apatosaurus louisae* holotype CM 3018 has a pneumatic foramen on its anterolateral surface (Gilmore 1936:plate 24). How did this happen?

As discussed by Witmer (1997:64), pneumatic diverticula can be viewed as “opportunistic pneumatizing machines, resorbing as much bone as possible within the constraints imposed by local biomechanical loading regimes”. As such, pneumatic variability routinely results in unexpected morphology with a great deal of individual variation. In fact diverticula are even more opportunistic than suggested here by Witmer, as they can make their way through soft-tissues as well as bone (Schachner et al. 2021). For example, pelicans and gannets have extensive subcutaneous diverticula (Richardson 1939, 1943, Daoust et al. 2008:figure 3). Soft-tissue diverticula similarly unmoored from major blood vessels occasionally find their way back to bone (O’Connor 2004), enabling pneumatic features to develop even in aspects of a bone where no major blood vessel enters.

Thus the seven-sites plan discussed herein provides a framework that accounts for the great majority of sauropod pneumaticity, but the opportunistic behavior of diverticula means that this schema cannot be exhaustive. It remains possible – and perhaps inevitable – that “out of bounds” pneumatic features will be found in places the model does not predict.

Pneumatic features in unusual neural canals

As described by Wedel and Atterholt (2023), in some aberrant sauropod vertebrae the neural canal is enclosed entirely either by the arch or by the centrum. Neural canals in these positions will still carry the usual segmental spinal arteries, and as they penetrate the walls of the neural canal they will still have the capacity to carry diverticula with them, creating pneumatic foramina in the roof, walls and floor of the canal. The only difference is that different elements of the vertebra would thereby be pneumatized: either all neural-canal pneumatic features penetrate the arch, or they all penetrate the centrum. In other words, the opportunistic behavior of the diverticula remains the same, treating bone as bone without concern for the identity of the bony element that happens to be at hand.

The mundanity of sauropods

Famously, sauropods attained masses an order of magnitude greater than other terrestrial animals, and much speculation has arisen regarding anatomical novelties that may have enabled this gigantism — e.g.

pharyngeal slits for respiration (Gale 1997), multiple accessory hearts in the neck (Choy and Altman 1992), and obligate aquatic habits (as overturned by Kermack 1951, Coombs 1975, Naish 2012 and others). Such ideas are entertaining but have no basis in fossilized evidence. Instead, the conservatism of the sauropod vascular system, as inferred from patterns of vertebral pneumaticity, provides evidence that they were built substantially like other animals, rather than being special magical monsters. Vascular and pneumatic foramina give us a window on the fact that, at the most basic level, sauropod skeletons developed much like those of extant amniotes.

Directions for future research

After roughly a century of only sporadic attention to dinosaurian pneumaticity (e.g., Janensch 1947), work by Witmer (1987, 1990, 1995, 1997), Britt (1993, 1997) and Britt et al. (1998) kicked off a broad resurgence of interest in both cranial and postcranial pneumaticity in dinosaurs and pterosaurs. After the past quarter century of intense work, we now find ourselves in the curious position of knowing far more about pneumaticity in dinosaurs than we do about the underlying vasculature. A systematic survey of vascular foramina in even in a single nearly-complete dinosaur skeleton, like the work done by Smuts (1974, 1975, 1976, 1977a, 1977b) on cattle, would be a welcome advance.

In this paper we have documented that sites of vertebral pneumatization in sauropods correspond with sites of vertebral vascularization in amniotes generally. Further, we have proposed that this correspondence represents causation rather than mere correlation – pneumatic diverticula invaded vertebrae and ribs in predictable locations because they followed pre-existing blood vessels as they developed. This hypothesis can be tested by looking for counter-examples to our seven-sites model. The occasional exception, like pneumatization of the outer surface of a dorsal rib in *Apatosaurus louisae*, can be chalked up to the variability and developmental opportunism of pneumatic diverticula (see Witmer 1997), but a large number of exceptions could force a reevaluation of our hypothesis, as could further experimental work on the actual process of postcranial pneumatization, following the example of Bremer (1940).

Furthermore, the seven-sites model could provide a framework for future investigations comparing the existence or prevalence of vascular and pneumatic traces serially within an individual, ontogenetically within a species or genus, and phylogenetically among lineages. This is an area in which little is known, but rapid progress could potentially be made simply by parsing existing published information using the seven sites as an organizational framework.

Like the collections surveys of Drumheller et al. (2020), McHugh et al. (2020, 2023) and McHugh and Drumheller (2026), all of our proposed investigations — documenting vascular foramina in the skeletons of dinosaurs (and other extinct vertebrates), searching for pneumatic features in places that might contradict the "blood vessels first" hypothesis, and using the seven sites as a framework for investigating serial, ontogenetic, and phylogenetic variation — are inherently collections-based projects that could leverage existing resources and involve students and early-career researchers; and they could be done in parallel, separately or collaboratively, across many subject taxa and many institutions. Even if we are mistaken about the morphogenetic basis of pneumaticity, we hope to inspire useful work.

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Figure Captions

Figure 1. Two schematic sauropod vertebrae, showing the homologous ossifications that compose them. **Left:** dorsal vertebra and free ribs. **Right:** caudal vertebra including fused ribs. Note that there are no haemal arches in precaudal (cervical, dorsal and sacral) vertebrae. Image composed from dorsal vertebra 4 of *Camarasaurus supremus* AMNH 5760'/D-X-131 in anterior view, modified from Osborn and Mook 1921:plate 70; left rib 4 (mirrored) of *Camarasaurus supremus* AMNH 5761/R-A-24 in anterior view, modified from Osborn and Mook 1921:figure 71; caudal vertebra 2 of *Camarasaurus lentus* (YPM collection, specimen number unknown) in anterior view, modified from Marsh 1896:plate 34, part 4; and haemal arch of *Camarasaurus grandis* (YPM collection, specimen number unknown) in posterior view, modified from Marsh 1896:plate 39: part 3c.

Figure 2. The major arteries and veins that serve each vertebral segment in amniotes. These vessels are largely conserved across Amniota; when a clade diverges from this pattern, it is usually through the loss of specific vessels, such as equatorial arteries in rabbits (Amato et al. 1959) and basivertebral veins in birds (O'Connor 2006).

Figure 3. The seven sites where arteries and veins enter and exit the bones (see Table 1).

Figure 4. Examples of vascular foramina in vertebrae of the American alligator, *Alligator mississippiensis*. **A.** The seven-site model from Fig. 3. **B.** Vascular foramina in the roof of the neural canal, going into the neural arch. **C1.** A vascular foramen on the lateral aspect of the neural arch. **C2.** A vascular foramen on the lateral aspect of the centrum. **D.** Vascular foramina in the floor of the neural canal, going into the centrum. **E.** A vascular foramen on the underside of the transverse process. **F.** A vascular foramen on the ventral aspect of the centrum. **G.** A vascular foramen on the medial aspect of the haemal arch. Specimens kindly provided by Andrew Farke.

Figure 5. Examples of vascular foramina in vertebrae of the Emperor penguin, *Aptenodytes forsteri*, LACM (Ornithology) 99854. **A.** The seven-site model from Fig. 3. **B.** Vascular foramina in the roof of the

neural canal, going into the neural arch. **C1.** Vascular foramina on the lateral aspect of the neural arch. **C2.** A vascular foramen on the lateral aspect of the centrum. **D.** Vascular foramina in the floor of the neural canal, going into the centrum. **E.** Vascular foramina on the underside of the transverse process. **F.** Vascular foramina on the ventral aspect of the centrum.

Figure 6. Examples of vascular foramina in the vertebrae of sauropods. **A.** The seven-site model from Fig. 3. **B.** Vascular foramina in the roof of the neural canal in an unfused cervical neural arch of *Brontosaurus* CM 555. **C1.** Vascular foramen on the lateral aspect of the neural arch in a proximal caudal of *Alamosaurus* USNM 15556. **C2.** A large vascular foramen on the lateral aspect of the centrum in a proximal caudal of *Alamosaurus* USNM 15556. **D.** Vascular foramina in the floor of the neural canal, going into the centrum in a caudal vertebra of cf. *Camarasaurus* FHPR 13443. **E.** A vascular foramen on the underside of the transverse process in a proximal caudal of *Alamosaurus* USNM 15556. **F.** A vascular foramen on the ventral aspect of the centrum in a caudal vertebra of cf. *Camarasaurus* FHPR 13443. G. Vascular foramina on the medial surface of a haemal arch of *Alamosaurus* USNM 15556.

Figure 7. Examples of pneumatic structures in sauropod vertebrae matching each of the seven sites of vascularization. **A.** The seven-site model from Fig. 3. **B.** Cervical neural arch of the titanosaur *Alamosaurus sanjuanensis* unnumbered BIBE specimen in ventral view (anterior to right), showing pneumatic fossae in the roof of the neural canal. Modified from slide 46 in supplementary information to Atterholt and Wedel (2018). **C.** Cervical vertebra 8 of the brachiosaurid *Giraffatitan brancai* paralectotype MB.R.2181, in left lateral view, showing extensive pneumatic sculpting in the neural spine and a deep pneumatic fossa in the side of the centrum. Photograph by authors. **D.** Dorsal vertebral centrum of the titanosaur *Mnyamawamtuka moyowamkia* holotype RRBP 05834 in dorsal view, anterior to top, showing pneumatic fossa in the floor of the neural canal. Modified from Gorscak and O'Connor (2019:figure 5C). **E.** Dorsal rib of the brachiosaurid *Brachiosaurus altithorax* holotype FMNH PR 25107, showing pneumatic foramen in the medial aspect. Modified from Riggs (1904:plate 75:figure 5). **F.** Cervical vertebra 6 of the brachiosaurid *Giraffatitan brancai* paralectotype MB.R.2181, in ventral view with anterior to the top, showing a distinct pneumatic foramen on the bottom of the centrum. Photograph by authors. **G.** haemal arch of indeterminate saltasaurine titanosaur MACN-Pv RN 233-4 in left posterolateral view, showing pneumatic foramen on medial aspect of right portion. Photograph courtesy of Virginia Zurriaguz.

Figure 1

Two schematic sauropod vertebrae, showing the homologous ossifications that compose them.

Left: dorsal vertebra and free ribs. Right: caudal vertebra including fused ribs. Note that there are no haemal arches in precaudal (cervical, dorsal and sacral) vertebrae. Image composed from dorsal vertebra 4 of *Camarasaurus supremus* AMNH 5760'/D-X-131 in anterior view, modified from Osborn and Mook 1921:plate 70; left rib 4 (mirrored) of *Camarasaurus supremus* AMNH 5761/R-A-24 in anterior view, modified from Osborn and Mook 1921:figure 71; caudal vertebra 2 of *Camarasaurus lentus* (YPM collection, specimen number unknown) in anterior view, modified from Marsh 1896:plate 34, part 4; and haemal arch of *Camarasaurus grandis* (YPM collection, specimen number unknown) in posterior view, modified from Marsh 1896:plate 39: part 3c.

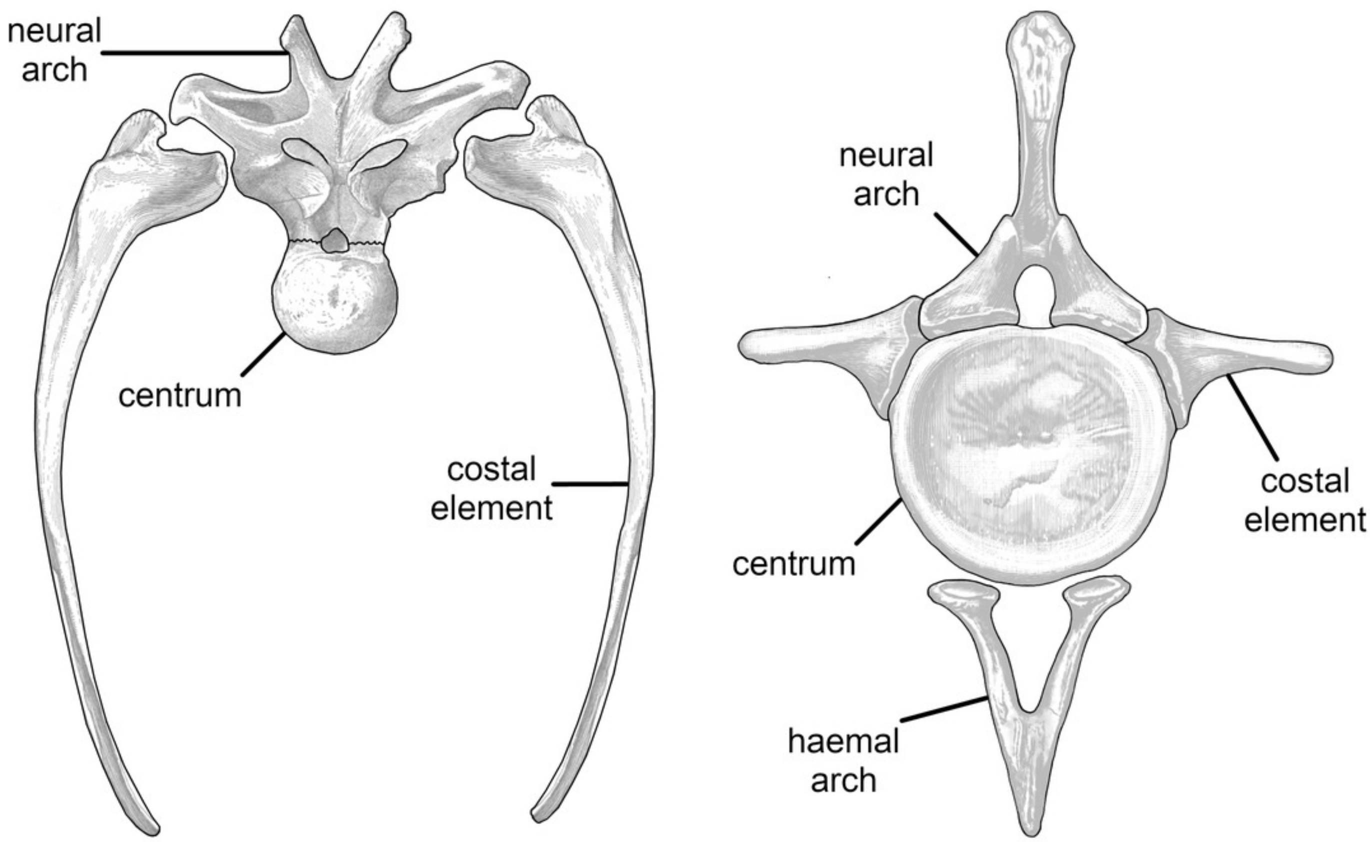
neural arch
centrum
costal element
neural arch
costal element
centrum
haemal arch

Figure 2

The major arteries and veins that serve each vertebral segment in amniotes.

These vessels are largely conserved across Amniota; when a clade diverges from this pattern, it is usually through the loss of specific vessels, such as equatorial arteries in rabbits (Amato et al. 1959) and basivertebral veins in birds (O'Connor 2006).

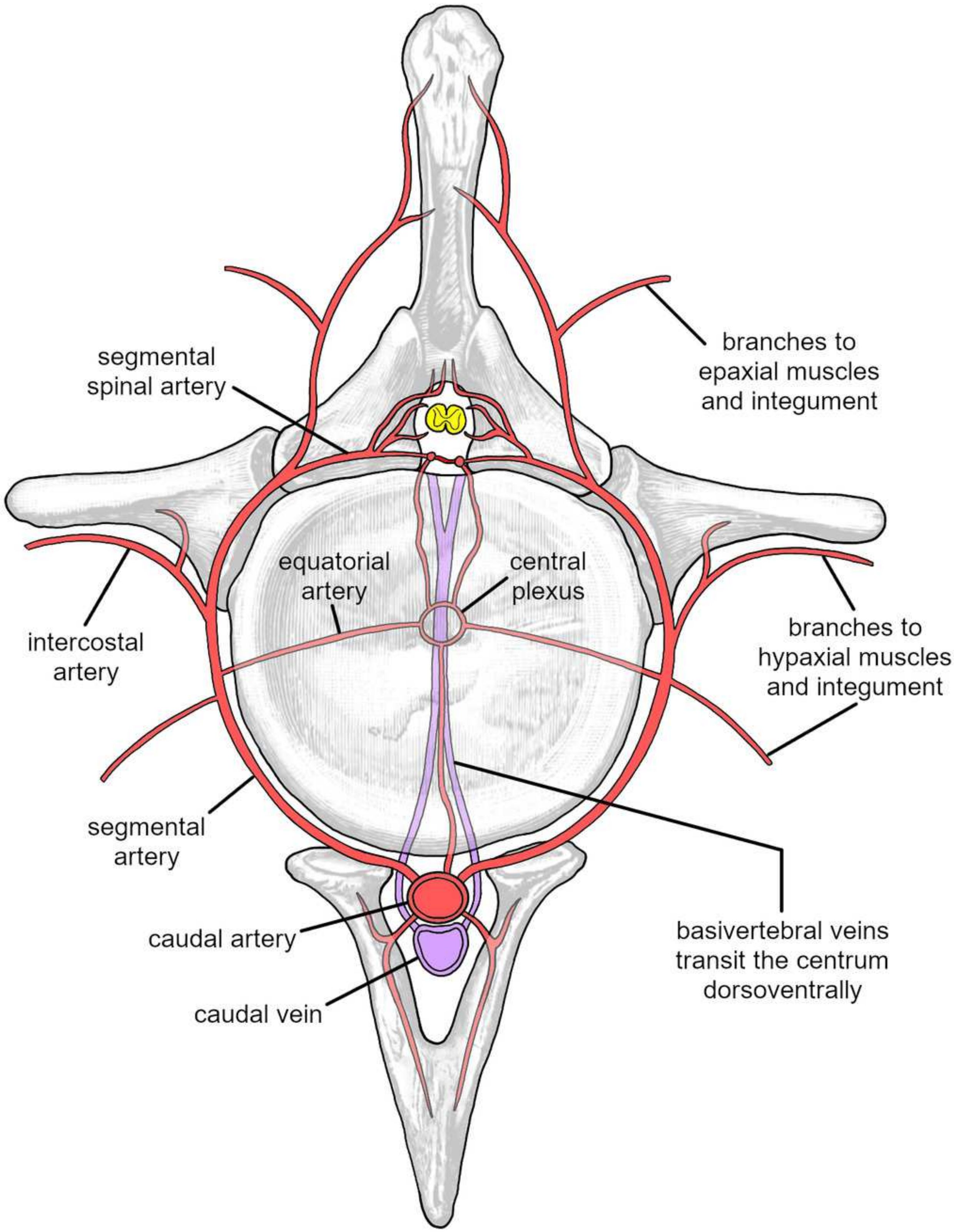
branches to epaxial muscles and integument
segmental spinal artery
equatorial artery
central plexus
intercostal artery
branches to hypaxial muscles and integument
segmental artery
caudal artery
caudal vein
basivertebral veins transit the centrum dorsoventrally

Figure 3

The seven sites where arteries and veins enter and exit the bones (see Table 1).

In life, most bones are surrounded by a dense vascular network, and microscopic vessels may enter and leave bones at many points. Only the most important vascularization sites are shown here.

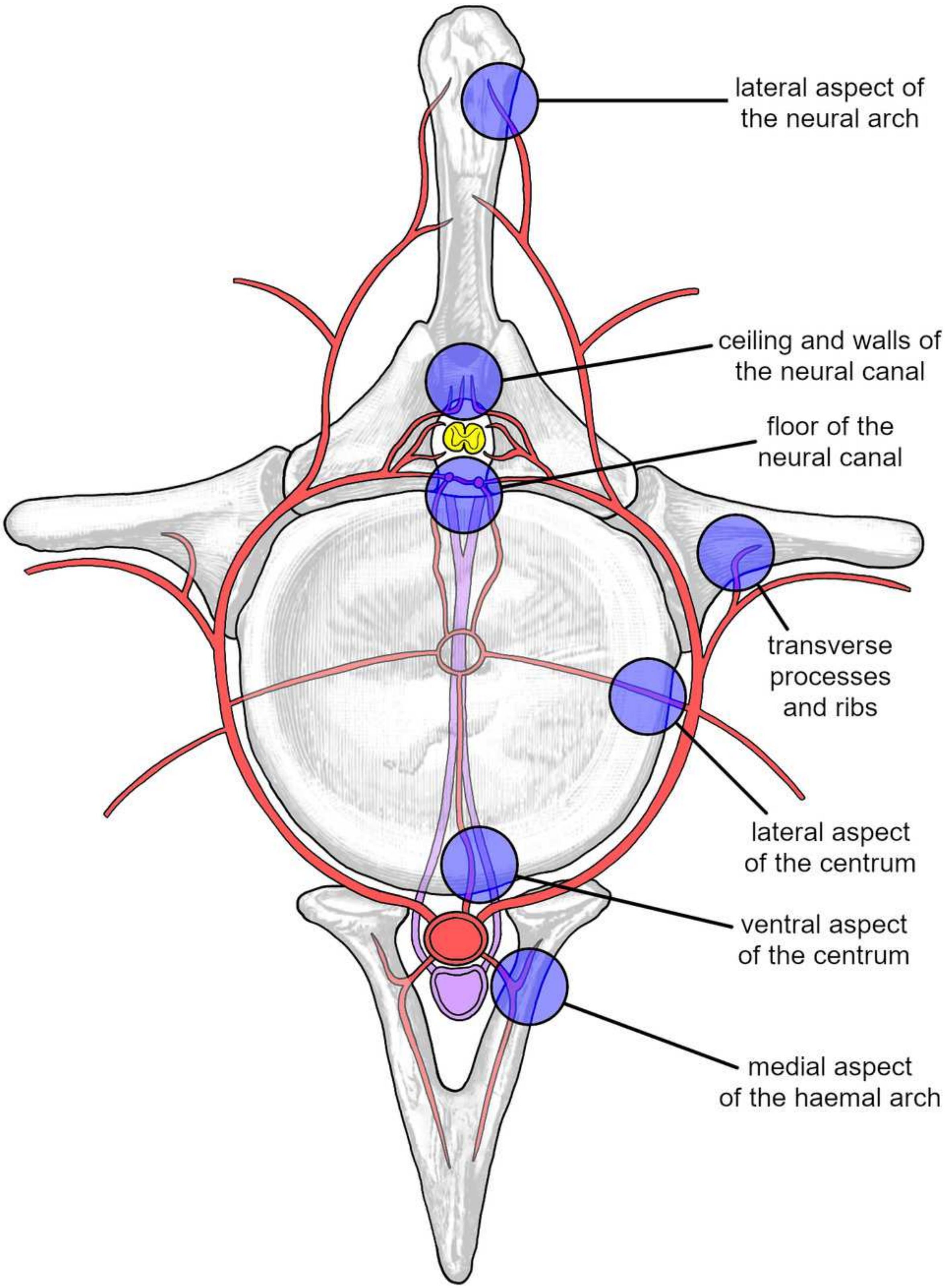
lateral aspect of
the neural arch
ceiling and walls of
the neural canal
floor of the
neural canal
transverse
processes
and ribs
lateral aspect
of the centrum
ventral aspect
of the centrum
medial aspect
of the haemal arch

Figure 4

Examples of vascular foramina in vertebrae of the American alligator, *Alligator mississippiensis*.

(A) The seven-site model from Fig. 3. (B) Vascular foramina in the roof of the neural canal, going into the neural arch. (C1) A vascular foramen on the lateral aspect of the neural arch. (C2) A vascular foramen on the lateral aspect of the centrum. (D) Vascular foramina in the floor of the neural canal, going into the centrum. (E) A vascular foramen on the underside of the transverse process. (F) A vascular foramen on the ventral aspect of the centrum. (G) A vascular foramen on the medial aspect of the haemal arch. Specimens kindly provided by Andrew Farke.

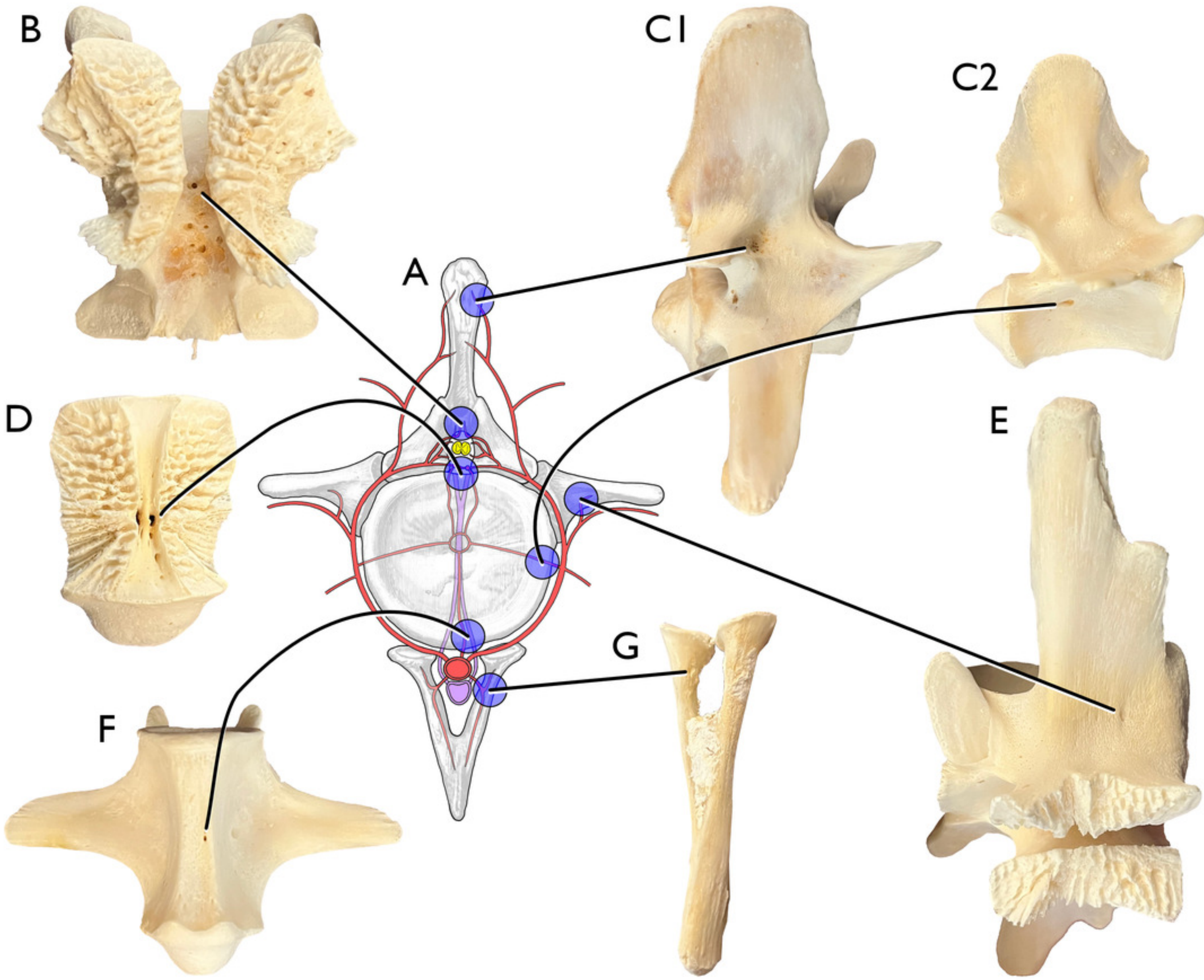
B
C1
C2
A
D
E
G
F

Figure 5

Examples of vascular foramina in vertebrae of the Emperor penguin, *Aptenodytes forsteri*, LACM (Ornithology) 99854.

(A) The seven-site model from Fig. 3. (B) Vascular foramina in the roof of the neural canal, going into the neural arch. (C1) Vascular foramina on the lateral aspect of the neural arch. (C2) A vascular foramen on the lateral aspect of the centrum. (D) Vascular foramina in the floor of the neural canal, going into the centrum. (E) Vascular foramina on the underside of the transverse process. (F) Vascular foramina on the ventral aspect of the centrum.

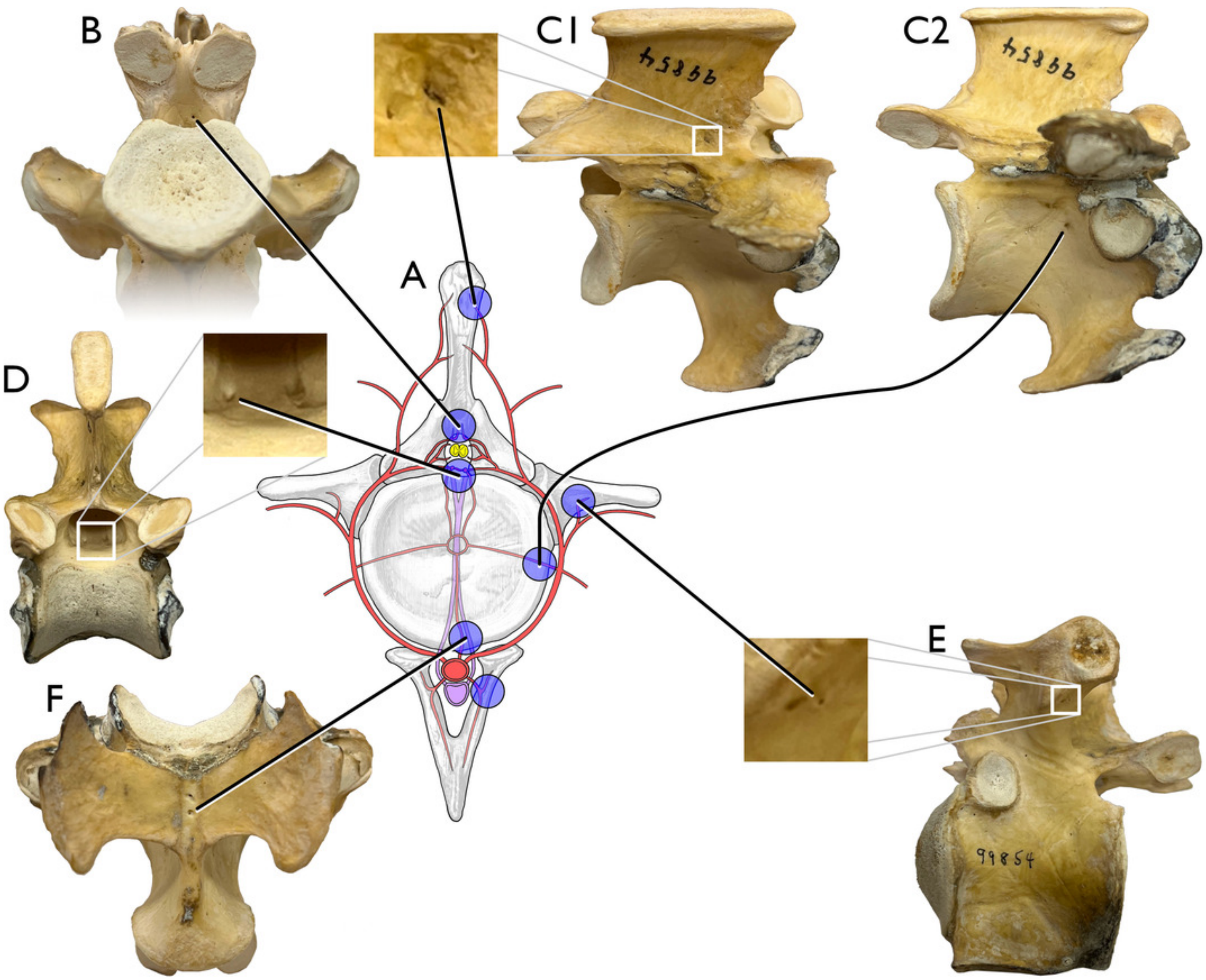
B
C1
C2
A
D
E
99854
F

Figure 6

Examples of vascular foramina in the vertebrae of sauropod dinosaurs.

(A) The seven-site model from Fig. 3. (B) Vascular foramina in the roof of the neural canal in an unfused cervical neural arch of *Brontosaurus* CM 555. (C1) Vascular foramen on the lateral aspect of the neural arch in a proximal caudal of *Alamosaurus* USNM 15556. (C2) A large vascular foramen on the lateral aspect of the centrum in a proximal caudal of *Alamosaurus* USNM 15556. (D) Vascular foramina in the floor of the neural canal, going into the centrum in a caudal vertebra of cf. *Camarasaurus* FHPR 13443. (E) A vascular foramen on the underside of the transverse process in a proximal caudal of *Alamosaurus* USNM 15556. (F) A vascular foramen on the ventral aspect of the centrum in a caudal vertebra of cf. *Camarasaurus* FHPR 13443. (G) Vascular foramina on the medial surface of a haemal arch of *Alamosaurus* USNM 15556.

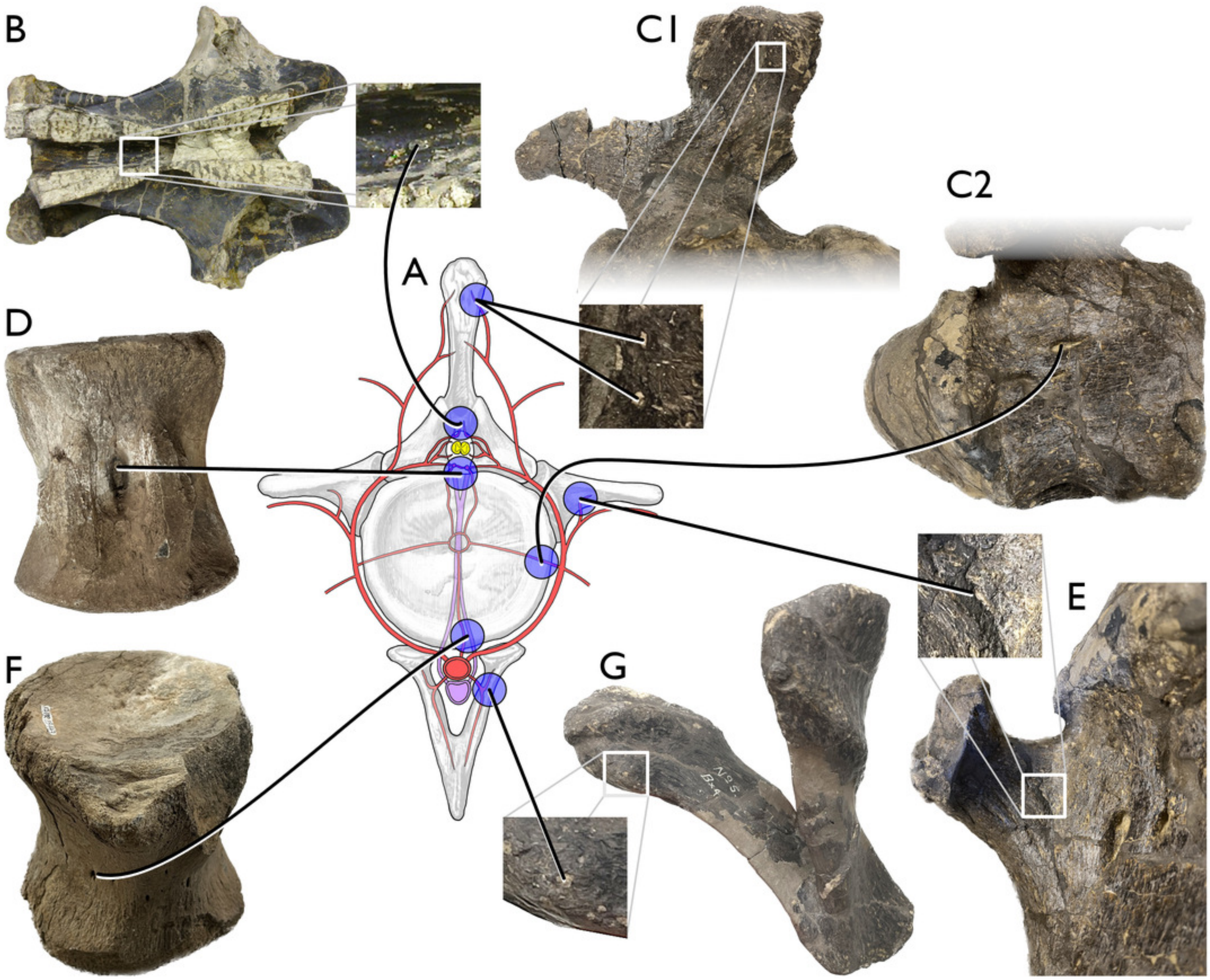
B
C1
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Figure 7

Examples of pneumatic structures in sauropod vertebrae matching each of the seven sites of vascularization.

(A) The seven-site model from Fig. 3. (B) Cervical neural arch of the titanosaur *Alamosaurus sanjuanensis* unnumbered BIBE specimen in ventral view (anterior to right), showing pneumatic fossae in the roof of the neural canal. Modified from slide 46 in supplementary information to Atterholt and Wedel (2018). (C) Cervical vertebra 8 of the brachiosaurid *Giraffatitan brancai* paralectotype MB.R.2181, in left lateral view, showing extensive pneumatic sculpting in the neural spine and a deep pneumatic fossa in the side of the centrum. Photograph by authors. (D) Dorsal vertebral centrum of the titanosaur *Mnyamawamtuka moyowamkia* holotype RRBP 05834 in dorsal view, anterior to top, showing pneumatic fossa in the floor of the neural canal. Modified from Gorscak and O'Connor (2019:figure 5C). (E) Dorsal rib of the brachiosaurid *Brachiosaurus altithorax* holotype FMNH PR 25107, showing pneumatic foramen in the medial aspect. Modified from Riggs (1904:plate 75:figure 5). (F) Cervical vertebra 6 of the brachiosaurid *Giraffatitan brancai* paralectotype MB.R.2181, in ventral view with anterior to the top, showing a distinct pneumatic foramen on the bottom of the centrum. Photograph by authors. (G) haemal arch of indeterminate saltasaurine titanosaur MACN-Pv RN 233-4 in left posterolateral view, showing pneumatic foramen on medial aspect of right portion. Photograph courtesy of Virginia Zurriaguz.

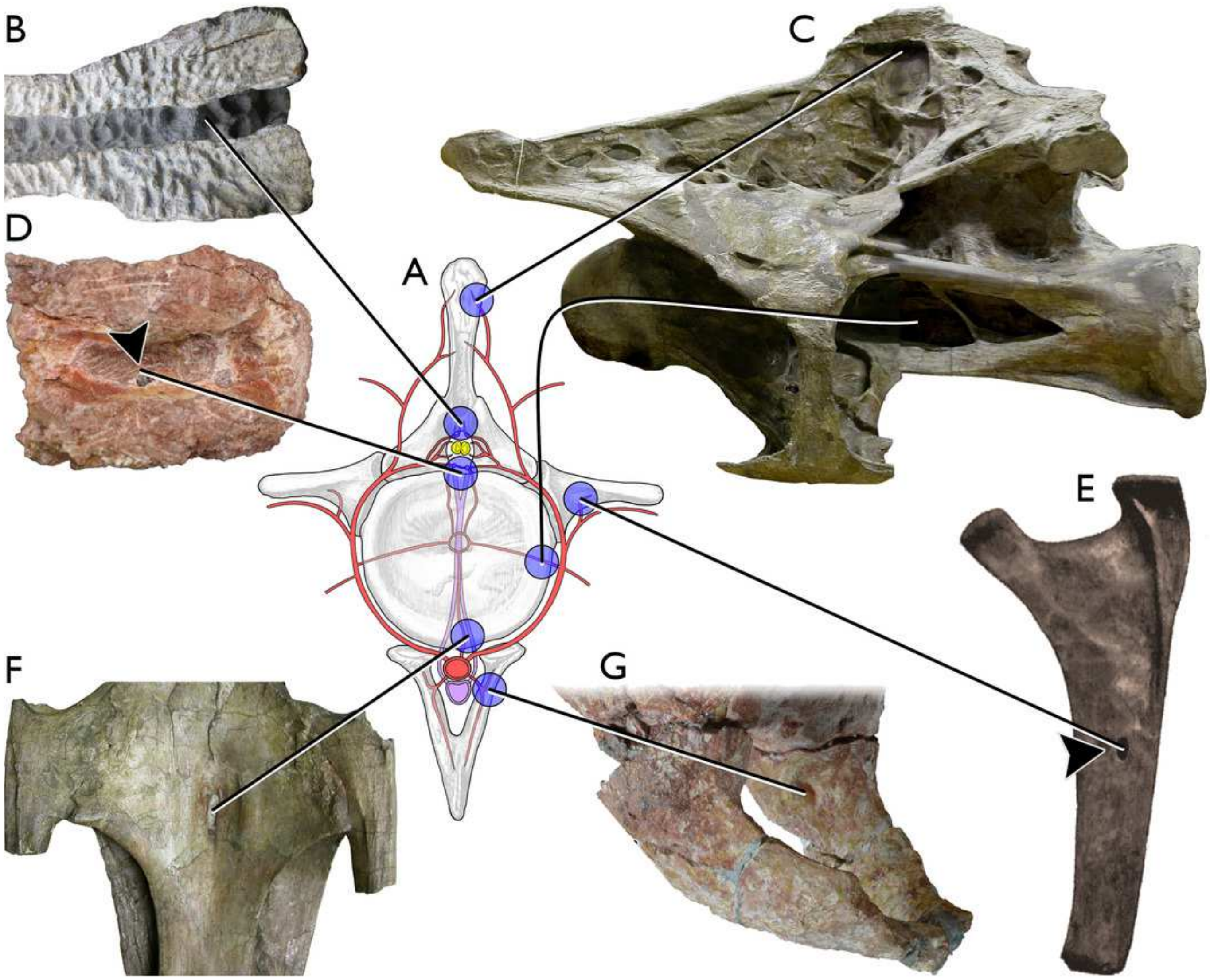
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Table 1(on next page)

Sites of vertebral vascularization and pneumaticity

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Table 1. Sites of vertebral vascularization and pneumaticity

#	Vascularized element	Vascularization site	Sauropod pneumaticity example
1	Neural arch	Lateral and anterior/posterior surfaces	Neural spine of 8th cervical vertebra of *Giraffatitan* (Figure 7C)
2	Neural arch	Dorsal roof or lateral walls of the neural canal	Neural canal roof of a cervical vertebrae of *Alamosaurus sanjuanensis* (Figure 7B)
3	Centrum	Ventral floor of the neural canal	Dorsal vertebral centrum of the titanosaur *Mnyamawamtuka moyowamkia* (Figure 7D)
4	Costal processes	Ventral surfaces of the transverse processes, and inner surfaces of the ribs	Medial face of dorsal rib of *Brachiosaurus altithorax* (Figure 7E)
5	Centrum	Lateral surfaces	Centrum of of 8th cervical vertebra of *Giraffatitan* (Figure 7C)
6	Centrum	Ventral surface	Ventral surface of cervical vertebra of *Giraffatitan* (Figure 7F)
7	haemal arch	Medial surfaces	Medial surface of a haemal arch of a saltasaurine titanosaur (Figure 7G)